Electric Vehicle Battery Protection System

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***Abstract*—- The electric vehicle drives through the battery; hence it is one of the most vital components of EVs. To ensure the security of battery from potential damages such as excessive charging & discharging and excessive overheating, the electric vehicle battery protection system (EV BPS) acts as a solution that helps in enhancing the reliability and lifespan of the battery. The EVBPS consists of three systems: the first system is battery management system (BMS) which is in charge of monitoring the state of charge, voltage, the second comes the charge control unit (CCU) for charging and discharging of the battery, and last but not least is the thermal management system (TMS). The TMS specifies a certain temperature range optimal for the battery for the prevention of the overheating or the fire hazards. To make the electric car safer and more dependable in its operation, its essential to have a system for battery management and monitoring with the battery pack.**

Keywords— Electric Vehicle Battery Protection System (EVBPS), Battery Management System (BMS), Charge Control Unit (CCU), Thermal Management System (TMS).

I. INTRODUCTION

The electric vehicles (EVs) that are advancing and are going to be future for a sustainable tomorrow also address the global problem of carbon emissions and global warming issues. The accelerating integration of EVs into our daily lives emphasizes the paramount experience of safeguarding the core vitality of the battery. However, the dynamic nature of batteries, coupled with the demands of vehicle operation, poses significant challenges in ensuring their longevity and reliability. Thus, the quest for an effective electric vehicle battery protection system (EVBPS) becomes imperative.

The EVs work with precision but its batteries are vulnerable to critical issues such charging issues of battery, overheating, cell unbalancing, thermal runaway, and fire hazards. As a solution to above concerns, to boost battery performance, including precise monitoring, charging- discharging control, heat management, battery safety, and protection, an effective BPS plays a significant role. The BPS emerges as the silent guardian, meticulously monitoring, regulating, and safeguarding the vitality of the battery pack.

This paper delivers a diverse review of different intelligent approaches and control arrangements for BPS in EV applications. In addition, in this review, the algorithms are assessed through battery state estimation concerning their features, structures, configuration, and accuracy.

The battery protection system comprises various components and algorithms designed for monitoring the safeguarding and controlling of the battery pack. This system will be effective when it comprises a BMS, which is central component for managing battery pack performance, safety, and longevity. The BMS comprises a battery pack in which individual battery cells are arranged in series and parallel configurations to provide the necessary voltage and energy storage capacity. The sensors are placed to measure parameters such as voltage, current, temperature, and state of charge, and the control unit processes data from sensors and implements control algorithms such as state of charge, state of health, cell balancing, and temperature management, etc. The cooling system regulates the temperature of the battery pack and the fuses or relays used to protect the battery pack from overcurrent, voltage, short circuits, and other electrical faults.

II. LITERATURE SURVEY

K. W. E. Cheng et al. [2] proposed a standard BMS block diagram with many functional blocks. The state of charge (SOC) calculation was performed by combining open-circuit voltage and Coulomb counting techniques. This removed the drawback of the standalone Coulomb counting approach. This included the modelling of the battery with SOC as one of the state variables; this estimate was further adjusted using the Kalman filtering technique.

John Chatzakis et al. [1] Modern BMS methodologies and a new layout method for a generalized reliable BMS was examined in this paper. It suggested primary advantages of BMS are its fault-tolerant capability and battery protection, when compared to the current systems. The suggested BMS comprised of several smart battery modules (SBMs), where each offers a series of battery cell, battery equalization, and protection. The theoretical expectations are confirmed by the experimental results of an assessment SBM that was created and tested in the laboratory.

Irsyad Nashirul Haq, Edi Leksono et al. [6] In this study, the design and testing of BMS cell monitoring and protection for lithium ferrophosphate (LFP) battery cells is presented. The functions of the proposed BMS cell monitoring and protection system, including data collection, safety protection, battery condition prediction and determination, and control over charging and discharging, have been shown. In the field of electric car R&D, an adaptable BMS is sufficient to safeguard various battery types.

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Vinodkumar Etacheri1, Rotem Marom et al. [3] The main features of Li-ion batteries are covered in this review, including the fundamental science underlying their operation, the most important components, anodes, cathodes, and electrolyte solutions, and significant future directions for the R&D of advanced Li-ion battery technologies for demanding applications such as electric vehicles and load-leveling. The two major energy challenges are: the transition of ground transport from ICE cars to electrical propulsion, or the use of EVs in place of ICE-powered vehicles.

Liu-Henke, X.; Scherler et al. [5] In this paper, the design of an EKF for SOC estimation was the main focus to illustrate the methodology. The EKF was created, tested, and refined; the HiL simulations were carried performed using a BMS prototype that they had produced.

Fang Li1, Yongjun Min et al. [12] In this work, a unique real-time voltage-based problem detection technique for lithium-ion batteries used in electric vehicles is presented. Initially, the voltage distribution of battery cells in electric cars is verified, and causes are examined. Moreover, cell defects were found using kurtosis for the first time. Defective cells in the battery pack were found using multidimensional scaling and density-based spatial clustering of applications with noise after the kurtosis-based method alarm.

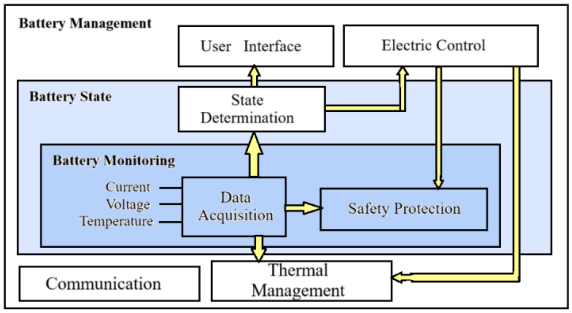
Mohamed et al. [9] provider an overview of the many wireless charging topologies available for use with electric vehicles. A brief overview of the various sectors' adoption of wireless charging technology is given. We discuss potential future uses for wireless power transmission (WPT), showing that placing WPT systems next to renewable energy systems is a feasible option.

Fuping Jiang, Xiaoming Xu et al. [8] To investigate the heat dissipation performance of various battery pack architectures, this study employs simulation approaches to focus on forced air cooling of battery packs. The findings show that, under the four different types of transient state conditions, both battery packs' rising temperatures are considerably higher than the temperature difference, and that a horizontal battery pack's maximum temperature rise, and temperature difference are smaller than those of a longitudinal battery pack. The temperature rise and temperature differential curves steepen as the electric car starts to slow down.

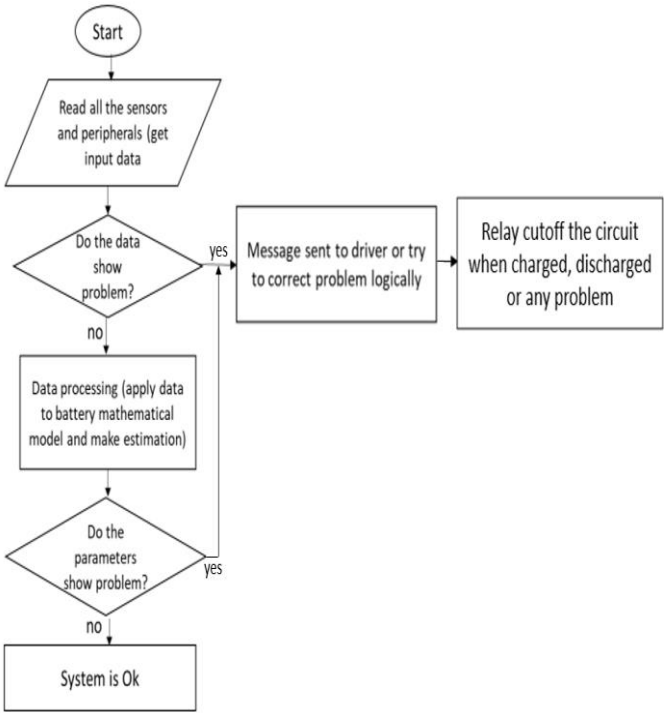
III. METHODOLOGY

Various types of battery protection systems are available for electric vehicles, including passive, active, and hybrid systems. The passive system depends on fuses and circuit breakers to protect the battery, whereas the active systems use electronic circuits to monitor and control the charging and discharging process of the battery. Hybrid systems combine the features of both passive and active systems to achieve optimal battery protection. The key parameters that affect the performance of an EVBPS include the battery's capacity, chemistry, and temperature range. An effective battery protection system must be able to manage these parameters efficiently to ensure about the safety and efficient operation of an electric vehicle.

The methodology for developing an electric vehicle (EV) battery protection system requires an understanding of the processes and the decision points involved in safeguarding the battery. The foremost step is requirement analysis which means identifying and understanding the specific requirements and constraints of the vehicle's battery system, that is, considering factors such as battery chemistry, voltage, capacity, and environmental conditions. Then, the technological evaluation in which the available technologies, components, and systems are assessed relevant to battery protection to assess the cost, scalability, and compatibility with the system. The roles and responsibilities of each component are defined and selected, such as sensors, actuators, controllers, and software tools, based on the system architecture and requirements. Then, the development of the software algorithms, which are the state of health (SOH), temperature management, fault detection, and emergency shutdown. The data that will be received from the sensors will be analysed to identify areas for improvement and optimisation, and the above information will be processed and hence communicated to the user.



The below flowchart represents the process flow for the battery protection system. As the process starts, that is, as the battery starts working and the data from the sensors attached to the battery as well as the peripherals are taken as the input data, in case of any problem, the message is sent to the controller. If the controller is able to rectify the error, the system will operate as it was, and the input data will be further processed according to the algorithms implied. After the result of the processed data, if the result does not show any error, then the system will function in its desired manner. In case of any error, the controller will direct the relay or the fuses to shut-down the system for the prevention of any hazards, which secures the battery as well as the vehicle.



IV. COMPONENT DESCRIPTION

Arduino- It is a32- bit ATmega328P microcontroller board, plays a major role in the implementation of a BPS for EVs. Battery voltage, temperature, and state of charge (SOC) can all be monitored in real time with the Arduino Uno because of its flexible digital and analogue input/output pins. To provide with the precise control over battery charging, and temperature regulation is made possible using Arduino Uno by interfacing with sensors and control circuits. The Arduino can be driven by using its IDE, providing with a programming platform which is easy-to-use and integrate into the system, and its USB interface facilitates communication with the system's onboard systems. It operates at 5 V voltage, and power supply is used when it is not connected through USB.



Figure 1- Arduino Uno

Buzzer: The piezoelectric buzzer is used in EV BPS, which provides an audible alert. Using the piezoelectric effect, in which a buzzer sound is produced due to electric charge which is produced by mechanical stress. In real- time, piezoelectric buzzer sounds an alert under certain conditions such as charging issues or overheating of the battery pack. The BPS functions are more effective because of the distinct sound produced by the buzzer. The buzzer is connected to a push button to limit duration of its sound. The buzzer's sound output ranges from 80 dB to 120 dB and frequency 2.5 kHz to 4 kHz.

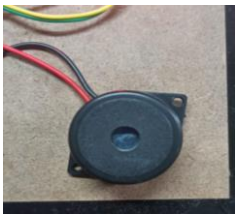


Figure 2- Buzzer

36 V Battery Pack- In the figure below is the 36 V battery pack, individual cells connected in series configuration to receive 36 V as output. EV's propulsion and overall functioning is due to the battery pack. It acts as an energy source to drive the vehicle and provide right voltage level to the EV's motor and auxiliary systems. The BMS is associated with battery pack only to keep track about the battery performance.



Figure 3- 36V Battery Pack

DC Motor- Fig 4 shows the DC motor acts as the main motor to create propulsion for the vehicle. The DC motor converts electrical energy into mechanical motion, through the energy drawn from the battery, propelling the vehicle forward. The DC motor optimises performance with BMS as it monitors the motor's power consumption, ensuring efficient energy use and avoiding problems of overcurrent and overheating. integrated into the EV BPS,



Figure 4- DC Motor

Voltage Divider as Voltage Sensor- Application of voltage sensor can be cleverly done through voltage divider in EV BPS. It is used for measuring high voltages. The voltage divider creates a fraction of total voltage across a specific point. A voltage reading is produced from the voltage signal across the divider. This enables the system to monitor real-time voltage level and apply preventive actions, including regulating charging rates or setting off alarms, guaranteeing the safety and effective operation within the electric car.

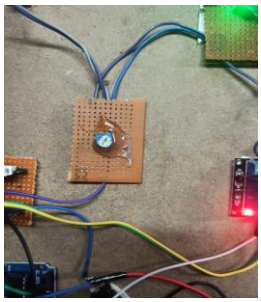


Figure 5- Voltage Divider

Relay- Fig 6 is the relay module, which is a switching device. This module's working is based on an electromagnetic relay, allowing rest of the components in battery system to be connected or isolated using a low-voltage signal from the BMS which activates a switchable to manage greater currents. Relays are major element incorporated in BMS to facilitate swift reactions to possible anomalies when the desired condition are not met. To defend the battery pack and the vehicle's general operation, the relay halts further damage by disconnecting the battery from the electrical system.

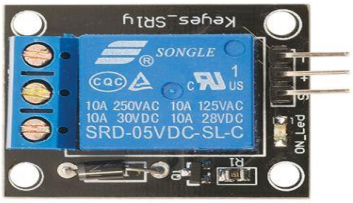


Figure 6- Relay Module

LCD- Fig. 7 is Arduino LCD, which offers a visual interface for tracking and display of desired data. The LCD's communication with BMS is done via Arduino platform. It displays real-time data which comprises of battery voltage, temperature, charging, and status. The navigable display enables users to rapidly evaluate the battery pack's condition. This improves the system's accessibility and user interfaces for setting protective by the BMS.



Figure 7- LCD

Transformer- Fig 8 shows a step-down transformer used in this project that is responsible to convert higher voltages to lower (here 220 Volts to 36 Volts). The power efficiency is maintained by lowering the voltage via electromagnetic induction. It's generally used in applications like electrical gadgets or charging systems. Proper voltage level is received by the electrical equipment. The step-down transformer reduces the risk of damage and improves the general safety and dependability of the linked devices.



Figure 8- Transformer

Rectifier: The rectifier is applied to shield the battery pack from undesirable voltage fluctuations & maximize charging efficiency, by converting the AC to DC. AC voltage produced by power source are converted into a format used to charge the battery pack is rectification. For the steadier voltage source, the rectifier utilizes the diodes for the current to flow in single direction & capacitor to smoothen the pulsating DC output.

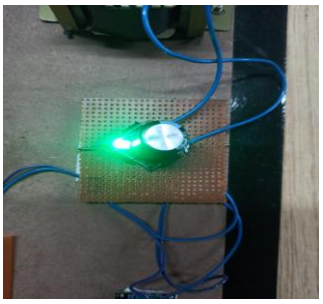


Figure 9- Rectifier

Current Sensor- This element of EV BPS, detects the electric current flow inside the battery pack precisely. The current sensor keeps an eye on the charging and discharging procedures. The sensor applies efficient protection techniques, preventing problems such as overcurrent that could cause damage or shorten battery lifespan, by providing real-time data on the current levels.



Figure 10- ACS712 Current Sensor

Temperature Sensor- The temperature of the battery pack is continuously monitored by a temperature sensor, which is a key element of EV BPS. In this sensor the safe temperature range for the battery pack is set, so when the sensor will capture the real- time data of temperature and if any value increases or decreases from the range specified, then preventive actions by signaling alerts or initiating emergency procedures are taken. Hence, in BMS, this sensor reduces the risk of thermal runaway or overheating, which leads to overall safety and long life of the EV system.



Figure 11- Temperature Sensor

V. PROPOSED IMPLEMENTATION

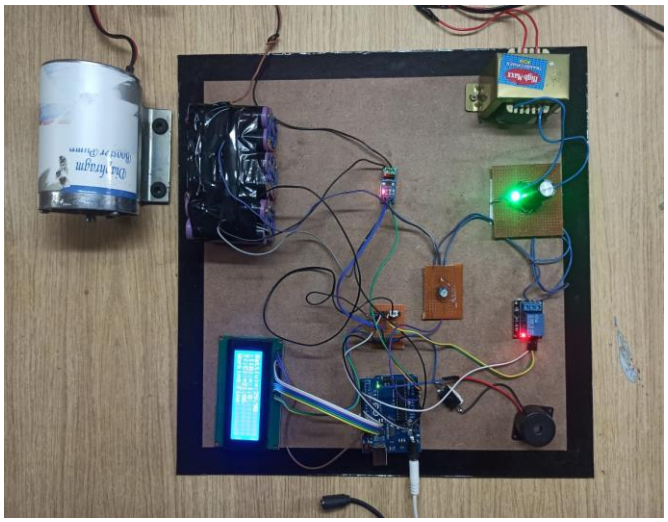


Figure 11- Electric vehicle battery protection system

In this paper, we have first implemented the battery pack of 36 Volts, and 18 cells each of 3.7 V were arranged in a manner to give an output of 36 Volt and maximum current. Three cells were combined in a series fashion, which would give an output of approximately 12 V (3.7 V * 3). Similarly, in a set of three, all cells were connected in series, and then another set of three cells connected in series were taken and joined parallel to the previous set. This was done to increase the output current. Thus, it became a set of six cells (two sets of three cells connected parallelly), which were then connected in series. Hence, we obtain an output of 4 amps and 36 V.

This proposed implementation seeks to create a robust and adaptive EV BPS that prioritises safety, efficiency, and long-term health of the battery pack in EV. This implementation involves association of various components and technologies to ensure the safety, and increase in span of the battery pack. This battery pack is charged using a charger, and the battery is drained using the DC motor, which draws power from the battery pack to convert electrical into mechanical energy, and pushes the vehicle in forward direction.

A step-down transformer and rectifier are employed to convert the 220-volt supply to a lower usable voltage (here 36V). The rectifier converts the AC power to DC, creating a regulated charging source & avoid undesirable fluctuations. This combination ensures for the battery to receive a stable and safe voltage.

Here, the Arduino is connected, and a connector is used to provide a uniform voltage of 5 V to every component in the circuit. The current passes through the current sensor (ACS712), which is used to monitor the flow of electric current into and out of the battery pack, which ensuring that the charge and discharge process remains within safe limits and prevents overcurrent situations. Then, the voltage divider, which is used as the voltage sensor, is used to measure the voltage level because it creates a fraction of the total voltage across a specific point. The voltage signal across the divider is fed into Arduino, which converts this signal into a useful voltage reading. The temperature sensor used here is a DS182B NTC thermistors, which are usually used for automobile purposes. This sensor measures the temperature of the individual cells or the overall pack. The data from this sensor helps the battery to operate in the optimal temperature range which is between 25ºC and 30ºC. The LCD is used to display real-time data of the voltage, current, and temperature and the status of the battery. When an anomaly is detected in the circuit or in real- time data, the buzzer will make a sound that can be controlled by using the push button. When the voltage is less than 30.5 V or greater than 35.8 V and the temperature exceeds the value of 30ºC along with the beep, the relay will cut off the connexions in the circuit, hence safeguarding the overall vehicle.

VI. RESULTS AND DISCUSSION

The intended aim was accomplished by the application of this system of EV BPS on the vehicle's battery.

First, the usage of battery monitoring and balancing feature made the batteries operate within certain specified limits and prevented the load on any single battery and distributed the charging or discharging evenly on each individual battery. The travel range of the EV has improved due to enhanced battery performance and lifespan.

Second, the mechanism for charge and discharge protection reduced the risk of battery charging problems and reduced the risk of thermal runaway and battery damage. This has not only secured the safety of electric cars but also reduced the risk of battery fires.

Then, with the help of the mechanism for temperature management, the batteries operate within a safe temperature range that averts overheating, which may have harmed the battery and its lifespan. In addition to ensuring optimal performance of the cooling system, it helped to the reduce energy consumption and enhance, the performance of the EV. Another essential element is the fault diagnostic function, which cuts the connexion and notifies the driver about battery problems. Hence, this system made an overall difference in the functioning and dependability of the battery.

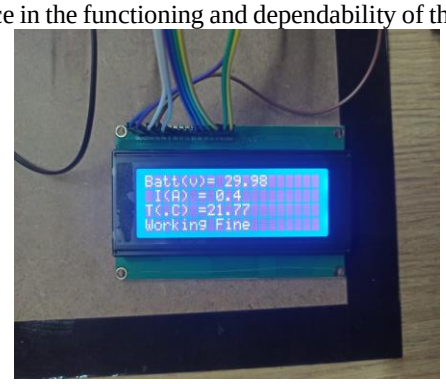


Figure 12- Displaying parameters (fine state)

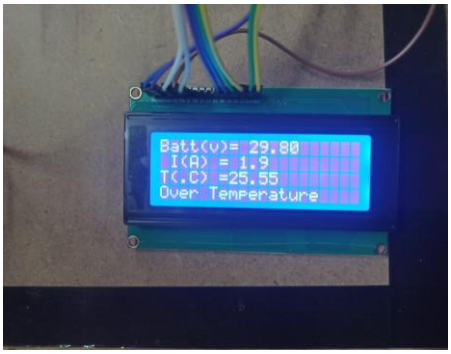


Figure 13- Fail state- High temperature detected.

VII. CONCLUSION

The development of the BPS for the electric vehicles was covered in this project. When integrated with the battery of the EVs, guarantees the security, dependability, and longevity of the battery pack. The union of the relays, sensors, algorithms manage and monitor the key parameters and the safety features vital for the battery such as voltage and current control, temperature management, fault detection, cell balancing, and fire hazards. The swift responses triggering any anomalies such as overheating and over- and under- charging contribute to prevention of any damage and hence defend the vehicle. Along with this, the added features like audible alerts, relays and display interfaces provides user with the comprehensive interaction and feedback with the battery. This has optimized EV's performance, promoting sustainable electric transportation.

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